

G2 Differentiating functions

Name: _____

Class: _____

Date: _____

Time: **495 minutes**

Marks: **410 marks**

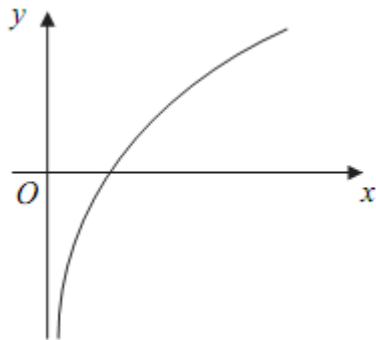
Comments:

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Q1.

The diagram shows the curve with equation $y = \ln(6x)$.



- (a) State the x -coordinate of the point of intersection of the curve with the x -axis.

(1)

- (b) Find $\frac{dy}{dx}$.

(2)

(Total 3 marks)

Q2.

- (a) (i) Find $\frac{dy}{dx}$ when $y = xe^{2x}$.

(3)

- (ii) Find an equation of the tangent to the curve $y = xe^{2x}$ at the point $(1, e^2)$

(2)

- (b) Given that $y = \frac{2 \sin 3x}{1 + \cos 3x}$, use the quotient rule to show that

$$\frac{dy}{dx} = \frac{k}{1 + \cos 3x}$$

where k is an integer.

(4)

(Total 9 marks)

Q3.

The depth of water, y metres, in a tank after time t hours is given by

$$y = \frac{1}{8}t^4 - 2t^2 + 4t, \quad 0 \leq t \leq 4$$

(a) Find:

(i) $\frac{dy}{dt}$,

(3)

(ii) $\frac{d^2y}{dt^2}$.

(2)

(b) Verify that y has a stationary value when $t = 2$ and determine whether it is a maximum value or a minimum value.

(4)

(c) (i) Find the rate of change of the depth of water, in metres per hour, when $t = 1$.

(2)

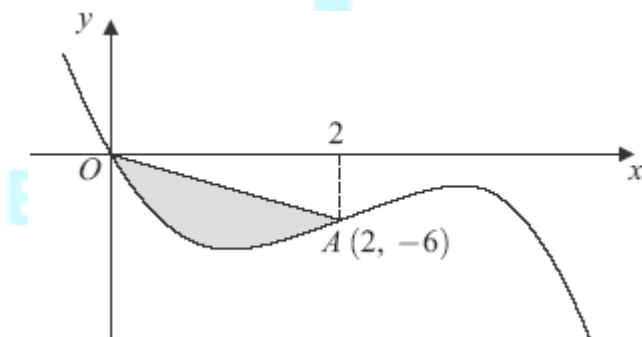
(ii) Hence determine, with a reason, whether the depth of water is increasing or decreasing when $t = 1$.

(1)

(Total 12 marks)

Q4.

The curve with equation $y = 12x^2 - 19x - 2x^3$ is sketched below.



The curve crosses the x -axis at the origin O , and the point $A(2, -6)$ lies on the curve.

(a) (i) Find the gradient of the curve with equation $y = 12x^2 - 19x - 2x^3$ at the point A .

(4)

(ii) Hence find the equation of the normal to the curve at the point A , giving your answer in the form $x + py + q = 0$, where p and q are integers.

(3)

(b) (i) Find the value of $\int_0^2 (12x^2 - 19x - 2x^3) dx$.

(5)

- (ii) Hence determine the area of the shaded region bounded by the curve and the line OA .

(3)

(Total 15 marks)

Q5.

A curve has equation $y = \frac{1}{x^3} + 48x$.

- (a) Find $\frac{dy}{dx}$.

(3)

- (b) Hence find the equation of each of the two tangents to the curve that are parallel to the x -axis.

(4)

- (c) Find an equation of the normal to the curve at the point $(1, 49)$.

(3)

(Total 10 marks)

Q6.

A curve has equation $y = e^{-4x}(x^2 + 2x - 2)$.

- (a) Show that $\frac{dy}{dx} = 2e^{-4x}(5 - 3x - 2x^2)$.

(3)

- (b) Find the exact values of the coordinates of the stationary points of the curve.

(5)

(Total 8 marks)

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Q7.

- (a) (i) Express $\sin 2\theta$ and $\cos 2\theta$ in terms of $\sin \theta$ and $\cos \theta$.

(2)

- (ii) Given that $0 < \theta < \frac{\pi}{2}$ and $\cos \theta = \frac{3}{5}$, show that $\sin 2\theta = \frac{24}{25}$ and find the value of $\cos 2\theta$.

(2)

- (b) A curve has parametric equations

$$x = 3 \sin 2\theta, \quad y = 4 \cos 2\theta$$

- (i) Find $\frac{dy}{dx}$ in terms of θ .

(3)

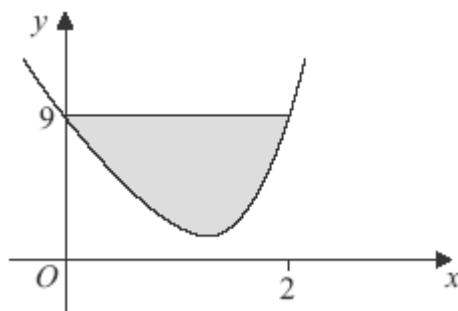
- (ii) At the point P on the curve, $\cos \theta = \frac{3}{5}$ and $0 < \theta < \frac{\pi}{2}$. Find an equation of the tangent to the curve at the point P .

(3)

(Total 10 marks)

Q8.

The curve with equation $y = x^4 - 8x + 9$ is sketched below.



The point $(2, 9)$ lies on the curve.

- (a) (i) Find $\int_0^2 (x^4 - 8x + 9) dx$.

(5)

- (ii) Hence find the area of the shaded region bounded by the curve and the line $y = 9$.

(2)

- (b) The point $A(1, 2)$ lies on the curve with equation $y = x^4 - 8x + 9$.

- (i) Find the gradient of the curve at the point A .

(4)

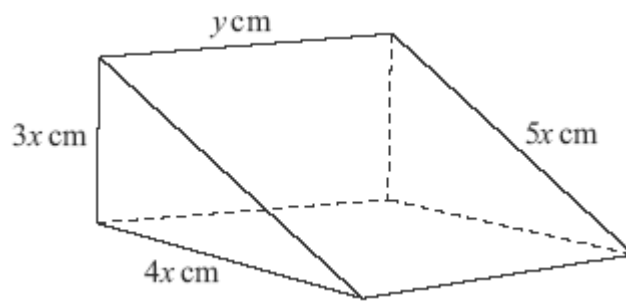
- (ii) Hence find an equation of the tangent to the curve at the point A .

(1)

(Total 12 marks)

Q9.

The diagram shows a block of wood in the shape of a prism with triangular cross-section. The end faces are right-angled triangles with sides of lengths $3x$ cm, $4x$ cm and $5x$ cm, and the length of the prism is y cm, as shown in the diagram.



The total surface area of the five faces is 144 cm^2 .

- (a) (i) Show that $xy + x^2 = 12$. (3)

- (ii) Hence show that the volume of the block, $V \text{ cm}^3$, is given by

$$V = 72x - 6x^3 \quad (2)$$

- (b) (i) Find $\frac{dV}{dx}$. (2)

- (ii) Show that V has a stationary value when $x = 2$. (2)

- (c) Find $\frac{d^2V}{dx^2}$ and hence determine whether V has a maximum value or a minimum value when $x = 2$. (2)
- (Total 11 marks)

Q10.

A curve C has the equation

$$y = \frac{x^3 + \sqrt{x}}{x}, x > 0$$

- (a) Express $\frac{x^3 + \sqrt{x}}{x}$ in the form $x^p + x^q$. (3)

- (b) (i) Hence find $\frac{dy}{dx}$. (2)

- (ii) Find an equation of the normal to the curve C at the point on the curve where

$$x = 1.$$

(4)

(c) (i) Find $\frac{d^2y}{dx^2}$.

(2)

(ii) Hence deduce that the curve C has no maximum points.

(2)

(Total 13 marks)

Q11.

(a) Find $\frac{dy}{dx}$ when:

(i) $y = \ln(5x - 2)$;

(2)

(ii) $y = \sin 2x$.

(2)

(b) The functions f and g are defined with their respective domains by

$$f(x) = \ln(5x - 2), \quad \text{for real values of } x \text{ such that } x \geq \frac{1}{2}$$

$$g(x) = \sin 2x, \quad \text{for real values of } x \text{ in the interval } -\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$$

(i) Find the range of f .

(2)

(ii) Find an expression for $gf(x)$.

(1)

(iii) Solve the equation $gf(x) = 0$.

(3)

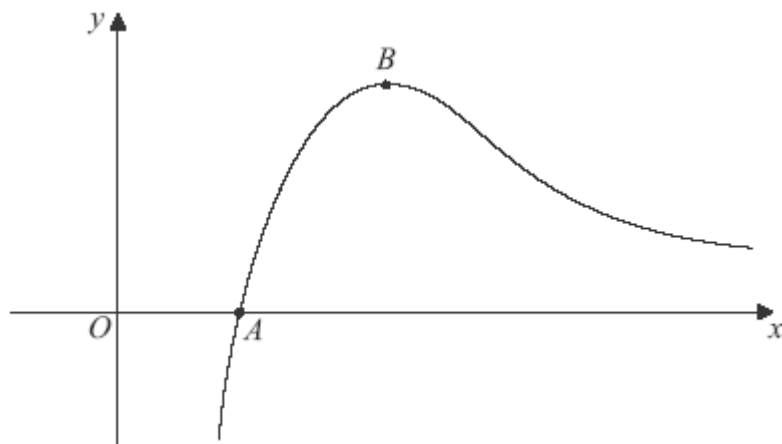
(iv) The inverse of g is g^{-1} . Find $g^{-1}(x)$.

(2)

(Total 12 marks)

Q12.

The diagram shows the curve $y = \frac{\ln x}{x}$.



The curve crosses the x -axis at A and has a stationary point at B .

- (a) State the coordinates of A . (1)
 - (b) Find the coordinates of the stationary point, B , of the curve, giving your answer in an exact form. (5)
 - (c) Find the exact value of the gradient of the normal to the curve at the point where $x = e^3$. (3)
- (Total 9 marks)

Q13.

A curve is defined by the parametric equations

$$x = 1 - 3t, \quad y = 1 + 2t^3$$

- (a) Find $\frac{dy}{dx}$ in terms of t . (3)
 - (b) Find an equation of the normal to the curve at the point where $t = 1$. (4)
 - (c) Find a cartesian equation of the curve. (2)
- (Total 9 marks)

Q14.

A model car moves so that its distance, x centimetres, from a fixed point O after time t seconds is given by

$$x = \frac{1}{2}t^4 - 20t^2 + 66t, \quad 0 \leq t \leq 4$$

(a) Find:

(i) $\frac{dx}{dt}$;

(3)

(ii) $\frac{d^2x}{dt^2}$.

(2)

(b) Verify that x has a stationary value when $t = 3$, and determine whether this stationary value is a maximum value or a minimum value.

(4)

(c) Find the rate of change of x with respect to t when $t = 1$.

(2)

(d) Determine whether the distance of the car from O is increasing or decreasing at the instant when $t = 2$.

(2)

(Total 13 marks)

Q15.

(a) The polynomial $p(x)$ is given by $p(x) = x^3 + x - 10$.

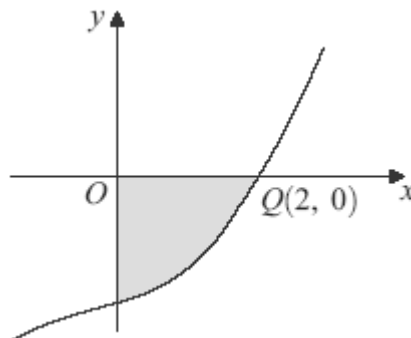
(i) Use the Factor Theorem to show that $x - 2$ is a factor of $p(x)$.

(2)

(ii) Express $p(x)$ in the form $(x - 2)(x^2 + ax + b)$, where a and b are constants.

(2)

(b) The curve C with equation $y = x^3 + x - 10$, sketched below, crosses the x -axis at the point $Q(2, 0)$.



(i) Find the gradient of the curve C at the point Q .

(4)

(ii) Hence find an equation of the tangent to the curve C at the point Q .

(2)

(iii) Find $\int (x^3 + x - 10) dx$.

(3)

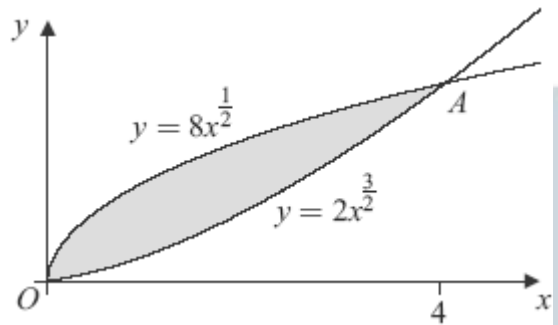
(iv) Hence find the area of the shaded region bounded by the curve C and the coordinate axes.

(2)

(Total 15 marks)

Q16.

The diagram shows a sketch of the curves with equations $y = 2x^{\frac{3}{2}}$ and $y = 8x^{\frac{1}{2}}$.



The curves intersect at the origin and at the point A , where $x = 4$.

(a) (i) For the curve $y = 2x^{\frac{3}{2}}$, find the value of $\frac{dy}{dx}$ when $x = 4$.

(2)

(ii) Find an equation of the normal to the curve $y = 2x^{\frac{3}{2}}$ at the point A .

(4)

(b) (i) Find $\int 8x^{\frac{1}{2}} dx$.

(2)

(ii) Find the area of the shaded region bounded by the two curves.

(4)

(c) Describe a single geometrical transformation that maps the graph of $y = 2x^{\frac{3}{2}}$

onto the graph of $y = 2(x+3)^{\frac{3}{2}}$.

(2)

(Total 14 marks)

Q17.

- (a) By using the binomial expansion, or otherwise, express $(1 + 2x)^4$ in the form

$$1 + ax + bx^2 + cx^3 + 16x^4$$

where a , b and c are integers.

(4)

- (b) Hence show that $(1 + 2x)^4 + (1 - 2x)^4 = 2 + 48x^2 + 32x^4$.

(3)

- (c) Hence show that the curve with equation

$$y = (1 + 2x)^4 + (1 - 2x)^4$$

has just one stationary point and state its coordinates.

(4)

(Total 11 marks)

Q18.

A curve has equation $y = e^{2x}(x^2 - 4x - 2)$.

- (a) Find the value of the x -coordinate of each of the stationary points of the curve.

(6)

- (b) (i) Find $\frac{d^2y}{dx^2}$.

(2)

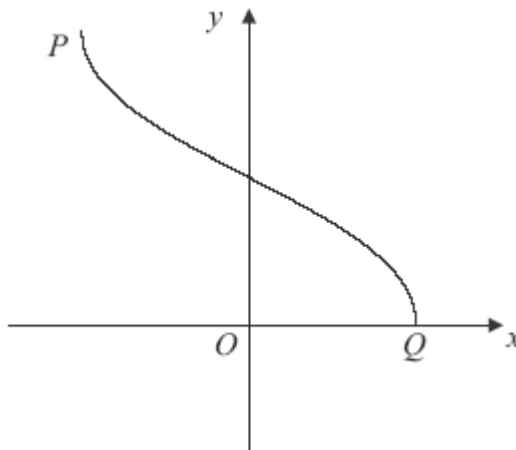
- (ii) Determine the nature of each of the stationary points of the curve.

(2)

(Total 10 marks)

Q19.

The sketch shows the graph of $y = \cos^{-1} x$.



- (a) Write down the coordinates of P and Q , the end points of the graph. (2)
- (b) Describe a sequence of two geometrical transformations that maps the graph of $y = \cos^{-1}x$ onto the graph of $y = 2 \cos^{-1}(x - 1)$. (4)
- (c) Sketch the graph of $y = 2 \cos^{-1}(x - 1)$. (2)
- (d) (i) Write the equation $y = 2 \cos^{-1}(x - 1)$ in the form $x = f(y)$. (2)
- (ii) Hence find the value of $\frac{dx}{dy}$ when $y = 2$. (3)

(Total 13 marks)

Q20.

- (a) Given that $y = \frac{4x}{4x-3}$, use the quotient rule to show that $\frac{dy}{dx} = \frac{k}{(4x-3)^2}$, where k is an integer. (2)
- (b) (i) Given that $y = x \ln(4x - 3)$, find $\frac{dy}{dx}$. (3)
- (ii) Find an equation of the tangent to the curve $y = x \ln(4x - 3)$ at the point where $x = 1$. (3)
- (c) (i) Use the substitution $u = 4x - 3$ to find $\int \frac{4x}{4x-3} dx$, giving your answer in

terms of x .

(4)

- (ii) By using integration by parts, or otherwise, find $\int \ln(4x - 3) dx$.

(4)

(Total 16 marks)

Q21.

The curve with equation $y = x^5 + 20x^2 - 8$ passes through the point P , where $x = -2$.

- (a) Find $\frac{dy}{dx}$.

(3)

- (b) Verify that the point P is a stationary point of the curve.

(2)

- (c) (i) Find the value of $\frac{d^2y}{dx^2}$ at the point P .

(3)

- (ii) Hence, or otherwise, determine whether P is a maximum point or a minimum point.

(1)

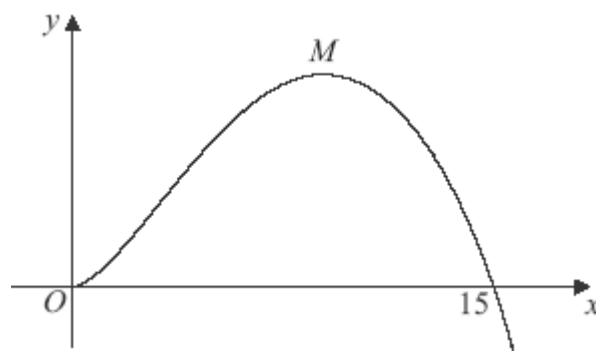
- (d) Find an equation of the tangent to the curve at the point where $x = 1$.

(4)

(Total 13 marks)

Q22.

The diagram shows part of a curve with a maximum point M .



The equation of the curve is

$$y = 15x^{\frac{3}{2}} - x^{\frac{5}{2}}$$

- (a) Find $\frac{dy}{dx}$. (3)
- (b) Hence find the coordinates of the maximum point M . (4)
- (c) The point $P(1, 14)$ lies on the curve. Show that the equation of the tangent to the curve at P is $y = 20x - 6$. (3)
- (d) The tangents to the curve at the points P and M intersect at the point R . Find the length of RM . (3)
- (Total 13 marks)

Q23.

- (a) The curve with equation

$$y = \frac{\cos x}{2x + 1}, \quad x > -\frac{1}{2}$$

intersects the line $y = \frac{1}{2}$ at the point where $x = \alpha$.

- (i) Show that α lies between 0 and $\frac{\pi}{2}$. (2)

- (ii) Show that the equation $\frac{\cos x}{2x + 1} = \frac{1}{2}$ can be rearranged into the form

$$x = \cos x - \frac{1}{2}$$

(1)

- (iii) Use the iteration $x_{n+1} = \cos x_n - \frac{1}{2}$ with $x_1 = 0$ to find x_3 , giving your answer to three decimal places. (2)

- (b) (i) Given that $y = \frac{\cos x}{2x + 1}$, use the quotient rule to find an expression for $\frac{dy}{dx}$. (3)

- (ii) Hence find the gradient of the normal to the curve $y = \frac{\cos x}{2x+1}$ at the point on the curve where $x = 0$.

(2)

(Total 10 marks)

Q24.

- (a) Use integration by parts to find $\int (t-1) \ln t \, dt$.

(4)

- (b) Use the substitution $t = 2x + 1$ to show that $\int 4x \ln(2x+1) \, dx$ can be written as $\int (t-1) \ln t \, dt$.

(3)

- (c) Hence find the exact value of $\int_0^1 4x \ln(2x+1) \, dx$.

(3)

(Total 10 marks)

Q25.

The curve with equation $y = x^4 - 32x + 5$ has a single stationary point, M .

- (a) Find $\frac{dy}{dx}$.

(3)

- (b) Hence find the x -coordinate of M .

(3)

- (c) (i) Find $\frac{d^2y}{dx^2}$.

(1)

- (ii) Hence, or otherwise, determine whether M is a maximum or a minimum point.

(2)

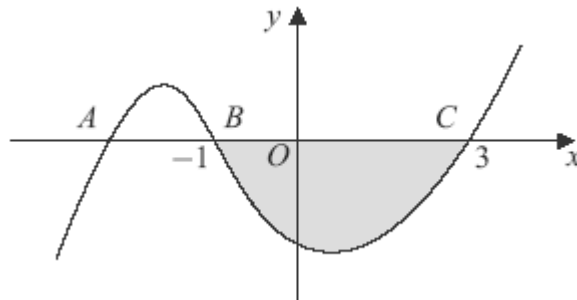
- (d) Determine whether the curve is increasing or decreasing at the point on the curve where $x = 0$.

(2)

(Total 11 marks)

Q26.

- (a) The polynomial $p(x)$ is given by $p(x) = x^3 - 7x - 6$.
- (i) Use the Factor Theorem to show that $x + 1$ is a factor of $p(x)$. (2)
- (ii) Express $p(x) = x^3 - 7x - 6$ as the product of three linear factors. (3)
- (b) The curve with equation $y = x^3 - 7x - 6$ is sketched below.



The curve cuts the x -axis at the point A and the points $B(-1, 0)$ and $C(3, 0)$.

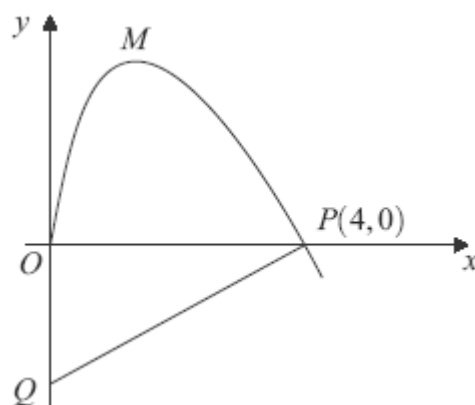
- (i) State the coordinates of the point A . (1)
- (ii) Find $\int_{-1}^3 (x^3 - 7x - 6) dx$. (5)
- (iii) Hence find the area of the shaded region bounded by the curve $y = x^3 - 7x - 6$ and the x -axis between B and C . (1)
- (iv) Find the gradient of the curve $y = x^3 - 7x - 6$ at the point B . (3)
- (v) Hence find an equation of the normal to the curve at the point B . (3)

(Total 18 marks)

Q27.

A curve, drawn from the origin O , crosses the x -axis at the point $P(4, 0)$.

The normal to the curve at P meets the y -axis at the point Q , as shown in the diagram.



The curve, defined for $x \geq 0$, has equation

$$y = 4x^{\frac{1}{2}} - x^{\frac{3}{2}}$$

- (a) (i) Find $\frac{dy}{dx}$. (3)
- (ii) Show that the gradient of the curve at $P(4, 0)$ is -2 . (2)
- (iii) Find an equation of the normal to the curve at $P(4, 0)$. (3)
- (iv) Find the y -coordinate of Q and hence find the area of triangle OPQ . (3)
- (v) The curve has a maximum point M . Find the x -coordinate of M . (3)
- (b) (i) Find $\int \left(4x^{\frac{1}{2}} - x^{\frac{3}{2}} \right) dx$. (3)
- (ii) Find the total area of the region bounded by the curve and the lines PQ and QO . (3)
- (Total 20 marks)**

Q28.

- (a) Find $\frac{dy}{dx}$ when:
- (i) $y = (2x^2 - 5x + 1)^{20}$;

(2)

(ii) $y = x \cos x.$

(2)

(b) Given that

$$y = \frac{x^3}{x-2}$$

show that

$$\frac{dy}{dx} = \frac{kx^2(x-3)}{(x-2)^2}$$

where k is a positive integer.

(3)

(Total 7 marks)

Q29.

Two numbers, x and y , are such that $3x + y = 9$, where $x \geq 0$ and $y \geq 0$.

It is given that $V = xy^2$.

(a) Show that $V = 81x - 54x^2 + 9x^3$.

(2)

(b) (i) Show that $\frac{dV}{dx} = k(x^2 - 4x + 3)$, and state the value of the integer k .

(4)

(ii) Hence find the two values of x for which $\frac{dV}{dx} = 0$.

(2)

(c) Find $\frac{d^2V}{dx^2}$.

(2)

(d) (i) Find the value of $\frac{d^2V}{dx^2}$ for each of the two values of x found in part (b)(ii).

(1)

(ii) Hence determine the value of x for which V has a maximum value.

(1)

(iii) Find the maximum value of V .

(1)
 (Total 13 marks)

Q30.

- (a) Write $\sqrt{x^3}$ in the form x^k , where k is a fraction.

(1)

- (b) A curve, defined for $x \geq 0$, has equation

$$y = x^2 - \sqrt{x^3}$$

- (i) Find $\frac{dy}{dx}$.

(3)

- (ii) Find the equation of the tangent to the curve at the point where $x = 4$, giving your answer in the form $y = mx + c$.

(5)

(Total 9 marks)

Q31.

Find $\frac{dy}{dx}$ when:

- (a) $y = (3x + 1)^5$;

(2)

- (b) $y = \ln(3x + 1)$;

(2)

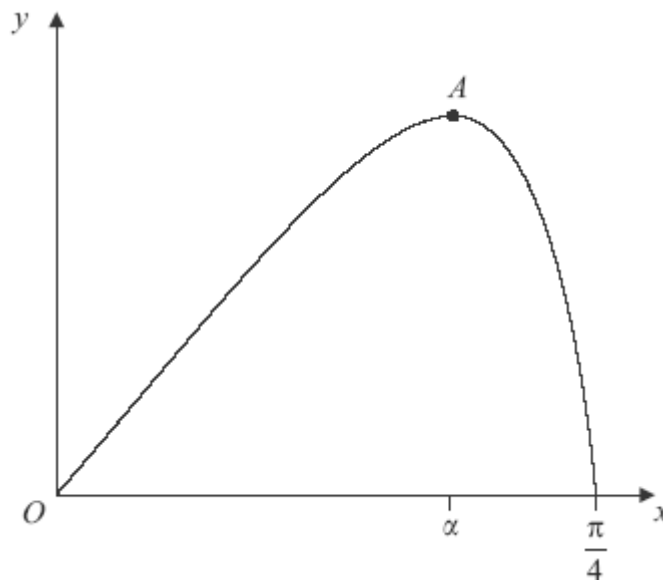
- (c) $y = (3x + 1)^5 \ln(3x + 1)$.

(3)

(Total 7 marks)

Q32.

A curve is defined for $0 \leq x \leq \frac{\pi}{4}$ by the equation $y = x \cos 2x$, and is sketched below.



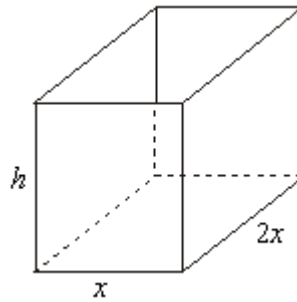
- (a) Find $\frac{dy}{dx}$. (2)
- (b) The point A , where $x = \alpha$, on the curve is a stationary point.
- (i) Show that $1 - 2\alpha \tan 2\alpha = 0$. (2)
- (ii) Show that $0.4 < \alpha < 0.5$. (2)
- (iii) Show that the equation $1 - 2x \tan 2x = 0$ can be rearranged to become $x = \frac{1}{2} \tan^{-1}\left(\frac{1}{2x}\right)$. (1)
- (iv) Use the iteration $x_{n+1} = \frac{1}{2} \tan^{-1}\left(\frac{1}{2x_n}\right)$ with $x_1 = 0.4$ to find x_3 , giving your answer to two significant figures. (2)
- (c) Use integration by parts to find $\int_0^{0.5} x \cos 2x dx$, giving your answer to three significant figures. (5)

(Total 14 marks)

Q33.

The diagram shows an **open-topped** water tank with a horizontal rectangular base and

four vertical faces. The base has width x metres and length $2x$ metres, and the height of the tank is h metres.



The combined internal surface area of the base and four vertical faces is 54 m^2 .

- (a) (i) Show that $x^2 + 3xh = 27$.

(2)

- (ii) Hence express h in terms of x .

(1)

- (iii) Hence show that the volume of water, $V \text{ m}^3$, that the tank can hold when full is given by

$$V = 18x - \frac{2x^3}{3}$$

(1)

- (b) (i) Find $\frac{dV}{dx}$.

(2)

- (ii) Verify that V has a stationary value when $x = 3$.

(2)

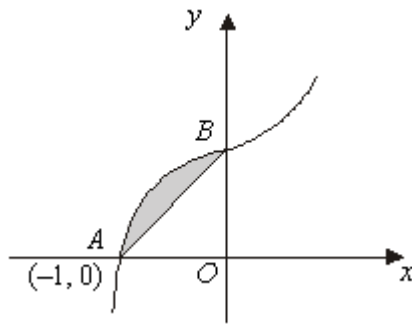
- (c) Find $\frac{d^2V}{dx^2}$ and hence determine whether V has a maximum value or a minimum value when $x = 3$.

(2)

(Total 10 marks)

Q34.

The curve with equation $y = 3x^5 + 2x + 5$ is sketched below.



The curve cuts the x -axis at the point $A(-1, 0)$ and cuts the y -axis at the point B .

- (a) (i) State the coordinates of the point B and hence find the area of the triangle AOB , where O is the origin.

(3)

- (ii) Find $\int (3x^5 + 2x + 5) dx$

(3)

- (iii) Hence find the area of the shaded region bounded by the curve and the line AB .

(4)

- (b) (i) Find the gradient of the curve with equation $y = 3x^5 + 2x + 5$ at the point $A(-1, 0)$.

(3)

- (ii) Hence find an equation of the tangent to the curve at the point A .

(1)

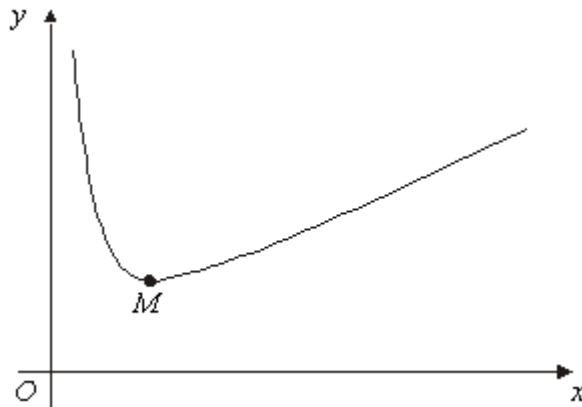
(Total 14 marks)

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Q35.

A curve C is defined for $x > 0$ by the equation $y = x + 1 + \frac{4}{x^2}$ and is sketched below.



- (a) (i) Given that $y = x + 1 + \frac{4}{x^2}$, find $\frac{dy}{dx}$. (3)
- (ii) The curve C has a minimum point M . Find the coordinates of M . (4)
- (iii) Find an equation of the normal to C at the point $(1, 6)$. (4)
- (b) (i) Find $\int \left(x + 1 + \frac{4}{x^2} \right) dx$. (3)
- (ii) Hence find the area of the region bounded by the curve C , the lines $x = 1$ and $x = 4$ and the x -axis. (2)

(Total 16 marks)